ECON 470 Homework 4

Submission 3, Spring 2026

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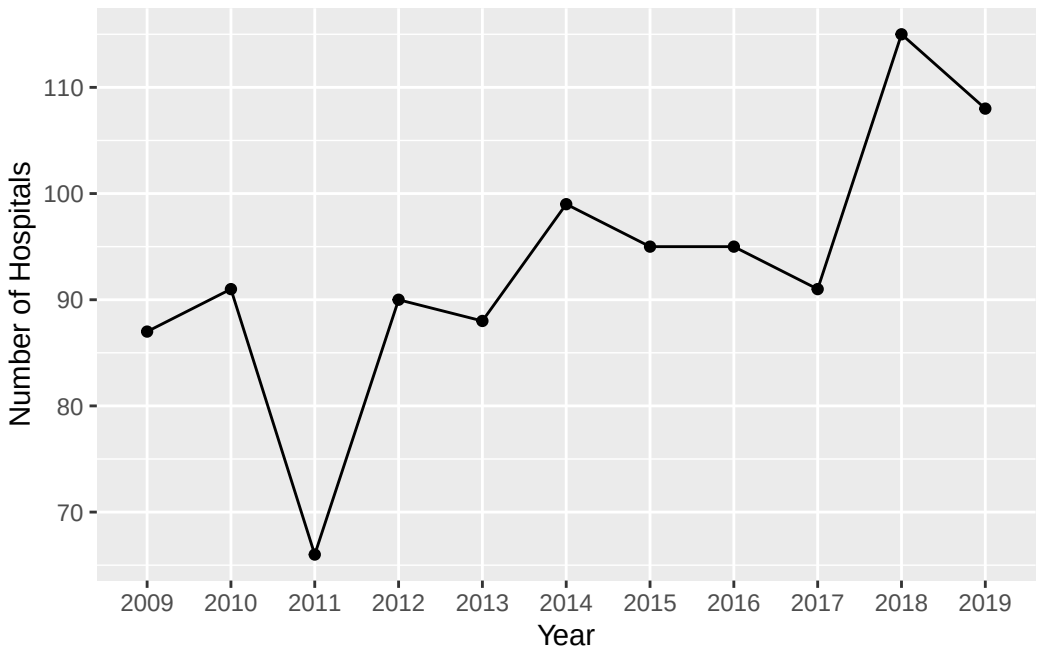
Data Cleaning

After creating the datasets, I calculated the estimated price using hospital cost report variables, and removed negative prices and observations in the bottom and top 1% to address outliers. I then constructed the data for questions 6-10. Price changes from 2011 to 2014 were calculated with simply subtracting the price from each year. Net penalty was a difference between hvbp and hrrp payment, and pre-period averages of Medicare discharges, bed size, and Medicaid discharges from 2009 to 2011 were also calculated.

Problem 1

The number of hospitals filing more than one report does not decline over time, suggesting that there is not a significant improvement


Number of Hospitals
110
100
90
80
70
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
Year

Problem 2

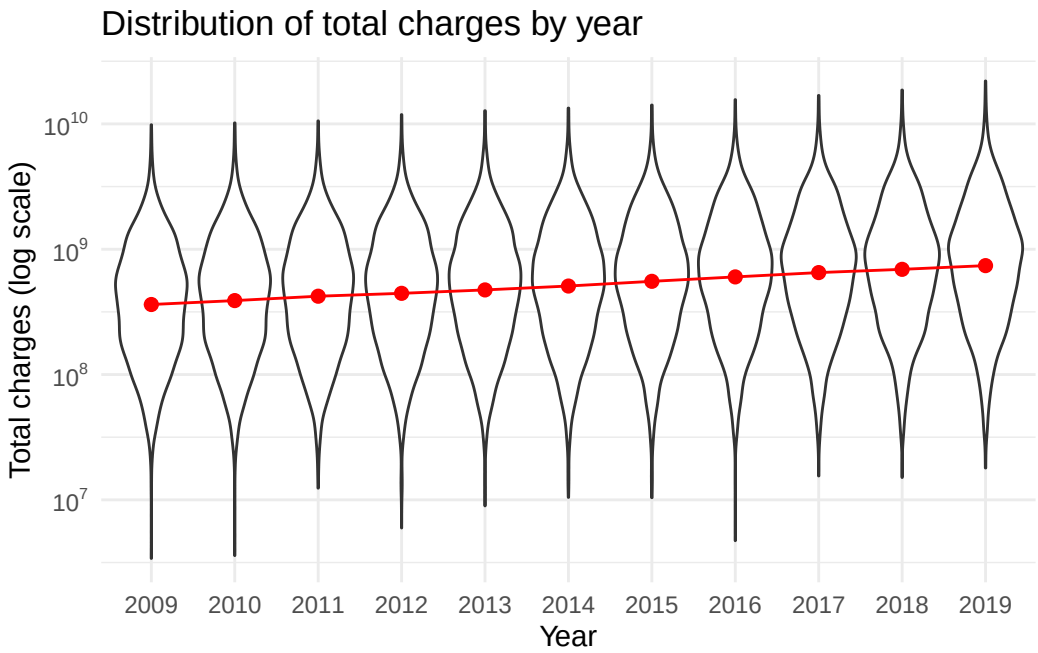
There are consistently around 7,000 unique hospital IDs per year.

Table 1: Unique Hospital IDs by Year

Year	Unique Hospitals
2009	6100
2010	6103
2011	6097
2012	6140
2013	6066
2014	6064
2015	6051
2016	6091
2017	6083
2018	6042
2019	6035

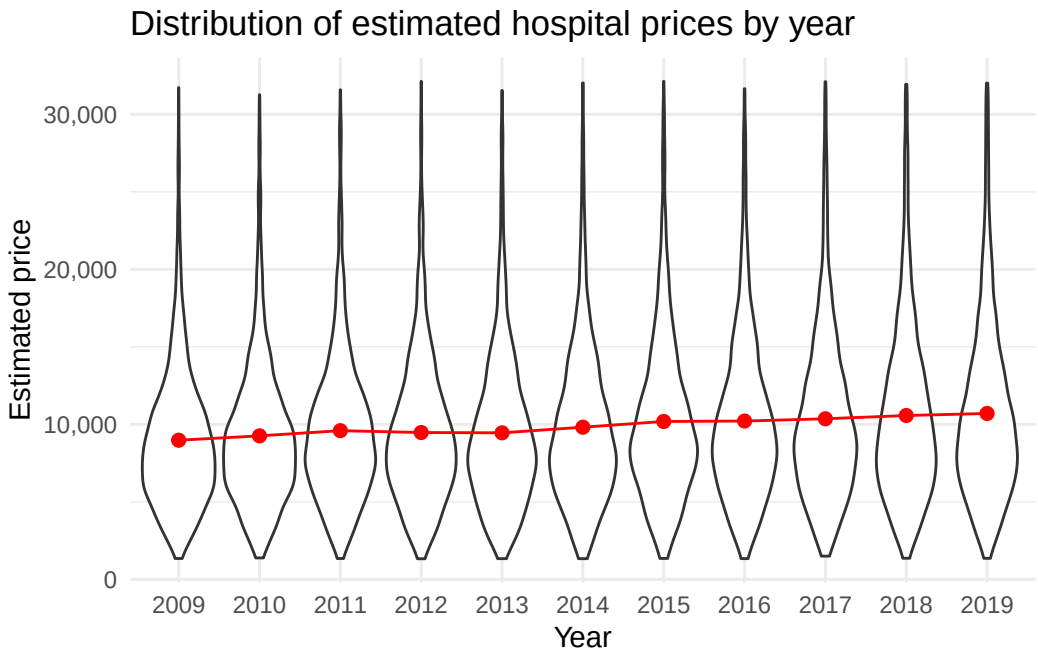
Problem 3

Total charges increased steadily from 2009-2019, which reflects growth in hospital billing across all hospital sizes.



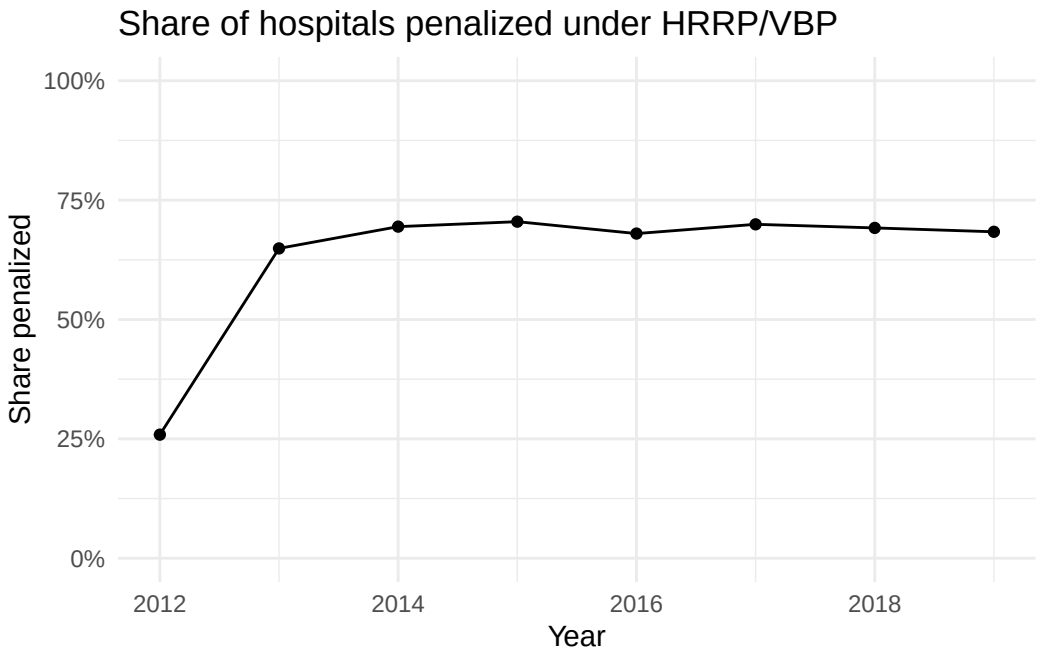
Problem 4

Estimated prices show an upward trend from 2009-2019, with a large variation across hospitals each year. The distribution centers around $10,000 per discharge.



Problem 5

The share of hospitals penalized under HRRP/VBP rose sharply from 2012 to 2013 and remained relatively stable around 70% through 2019. The sharp increase can be attributed to the HRRP/VBP being instilled from 2010 with the Affordable Care Act.



Problem 6

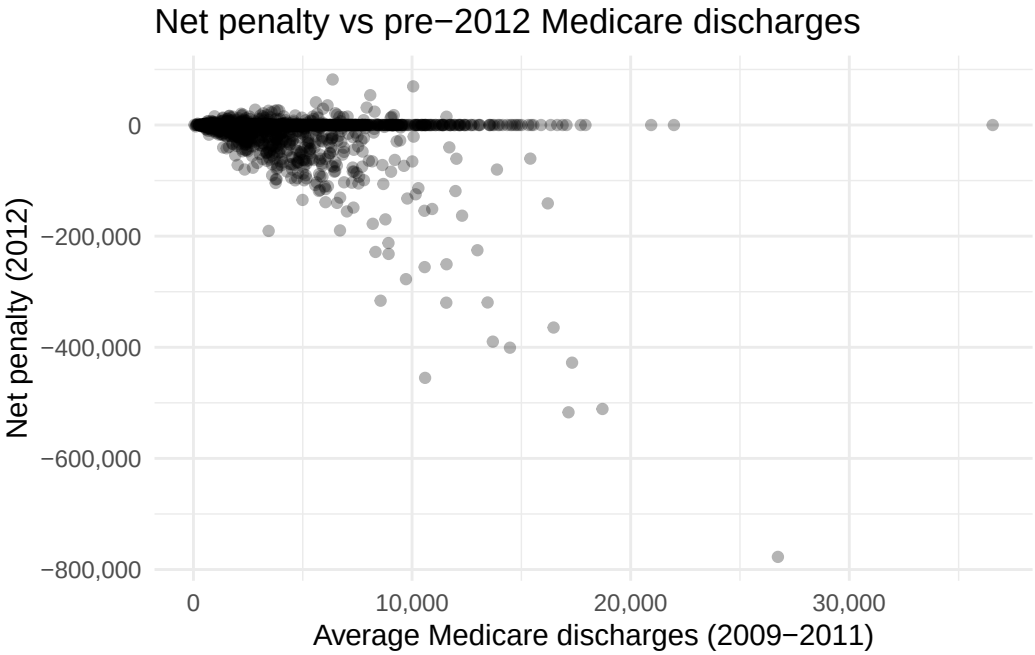
This table presents OLS estimates of the effect of net penalties on price changes across three different specifications. The coefficient on net penalty is small and negative across all specifications, which suggests slightly lower price growth in penalized hospitals, but this could be due to the endogeneity issue of penalty status.

Table 2: OLS Estimates of Net Penalty on Price Changes

Statistic	**OLS 1**	**OLS 2**	**OLS 3**
Coefficients			
Intercept	218.034	200.004	39.258
p-value	*0.000*	*0.000*	*0.565*
Net Penalty	-0.004	-0.004	-0.003
p-value	*0.007*	*0.009*	*0.031*
Mean Beds (Pre-2012)	0	0.074	0.065
p-value	*0*	*0.030*	*0.059*
Mean Medicaid Discharges (Pre-2012)	0	0	0.103
p-value	*0*	*0*	*0.000*
Model Fit			
# Obs	2395	2395	2395
R2	0.003	0.005	0.012

Problem 7

The scatterplot shows a negative relationship between net penalty and pre-2012 Medicare discharges. This suggests that hospitals with a larger proportion of Medicare patient tended to receive larger net penalties.



Problem 8

The first stage shows a strong negative relationship between pre-2012 Medicare discharges and net penalty. This confirms that Medicare discharge share is a good instrument. The reduced form shows a weaker but direct relationship between Medicare discharges and price changes.

Table 3: First Stage and Reduced Form Results

	First Stage			Reduced Form		
Statistic	**FS 1**	**FS 2**	**FS 3**	**RF 1**	**RF 2**	**RF 3**
Coefficients						
Intercept	6128.385	6126.161	5566.746	170.112	165.169	112.058
p-value	*0.000*	*0.000*	*0.000*	*0.039*	*0.045*	*0.177*
Avg Medicare	0	0	0	0	0	0
p-value	*0*	*0*	*0*	*0*	*0*	*0*
Mean Beds	0	0	0	0	0	0
p-value	*0*	*0*	*0*	*0*	*0*	*0*
Mean Medicaid	0	0	0	0	0	0
p-value	*0*	*0*	*0*	*0*	*0*	*0*
Model Fit						
# Obs	2395	2395	2395	2395	2395	2395
R2	0.115	0.115	0.119	0.001	0.003	0.01

Problem 9

The IV estimates show the local average treatment effect of net penalties for price changes in hospitals that had affected penalty status from Medicare volume.

```
TSLS estimation, Dep. Var.: price_change, Endo.: net_penalty, Instr.: mean_mcare
Second stage: Dep. Var.: price_change
Observations: 2,395
Standard-errors: IID
                  Estimate Std. Error  t value  Pr(>|t|)
(Intercept)     202.466600   65.75194  3.07925 0.0020988 **
fit_net_penalty  -0.005279    0.00400 -1.31971 0.1870594
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2,624.7   Adj. R2: 0.001976
F-test (1st stage), net_penalty: stat = 311.1     , p < 2.2e-16, on 1 and 2,393 DoF.
                     Wu-Hausman: stat =   0.189635, p = 0.66326, on 1 and 2,392 DoF.
```

```
TSLS estimation, Dep. Var.: price_change, Endo.: net_penalty, Instr.: mean_mcare
Second stage: Dep. Var.: price_change
Observations: 2,395
Standard-errors: IID
                  Estimate Std. Error  t value  Pr(>|t|)
(Intercept)     192.089233  65.875117  2.91596 0.0035789 **
fit_net_penalty  -0.004394   0.004016 -1.09416 0.2739946
mean_beds         0.073596   0.034336  2.14343 0.0321791 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2,621.5   Adj. R2: 0.003971
F-test (1st stage), net_penalty: stat = 308.0     , p < 2.2e-16 , on 1 and 2,392 DoF.
                     Wu-Hausman: stat =   0.050845, p = 0.821619, on 1 and 2,391 DoF.
```

```
TSLS estimation, Dep. Var.: price_change, Endo.: net_penalty, Instr.: mean_mcare
Second stage: Dep. Var.: price_change
Observations: 2,395
Standard-errors: IID
                 Estimate Std. Error t value   Pr(>|t|)
(Intercept)     85.270233  72.623492 1.17414 2.4046e-01
fit_net_penalty  0.004812   0.004188 1.14892 2.5070e-01
mean_beds        0.069548   0.034482 2.01693 4.3814e-02 *
mean_mcaid       0.118904   0.026819 4.43355 9.6885e-06 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2,629.9   Adj. R2: -0.00288
F-test (1st stage), net_penalty: stat = 286.2    , p < 2.2e-16 , on 1 and 2,391 DoF.
                     Wu-Hausman: stat =   3.88295, p = 0.048894, on 1 and 2,390 DoF.
```

```r
options(knitr.kable.NA = 0)
library(knitr)
library(kableExtra)

fmt_p <- function(p) ifelse(is.na(p), NA, sprintf("%.3f", p))

get_coef <- function(model, var) {
  ct <- coef(model)
  if (var %in% names(ct)) round(ct[var], 3) else NA
}
get_p <- function(model, var) {
  pt <- coeftable(model)[, "Pr(>|t|)"]
  if (var %in% names(pt)) fmt_p(pt[var]) else NA
}

data.frame(
  Statistic = c("Intercept", "p-value", "Net Penalty", "p-value", "Mean Beds", "p-value", "Me
  `IV 1` = c(get_coef(iv1, "(Intercept)"), get_p(iv1, "(Intercept)"), get_coef(iv1, "fit_net_
  `IV 2` = c(get_coef(iv2, "(Intercept)"), get_p(iv2, "(Intercept)"), get_coef(iv2, "fit_net_
  `IV 3` = c(get_coef(iv3, "(Intercept)"), get_p(iv3, "(Intercept)"), get_coef(iv3, "fit_net_
  check.names = FALSE
) %>%
  kable(
    col.names = c("Statistic", "IV 1", "IV 2", "IV 3"),
    digits  = 3,
    booktabs = TRUE,
    caption = "IV Results of Penalty Status on Price Change"
  ) %>%
  kable_styling(latex_options = c("hold_position", "striped")) %>%
  column_spec(1, bold = TRUE) %>%
  row_spec(0, bold = TRUE) %>%
  row_spec(c(2, 4, 6, 8), italic = TRUE, color = "gray") %>%
  pack_rows("Coefficients", 1, 8) %>%
  pack_rows("Model Fit", 9, 10)
```

Table 4: IV Results of Penalty Status on Price Change

Statistic	**IV 1**	**IV 2**	**IV 3**
Coefficients			
Intercept	202.467	192.089	85.27
p-value	*0.002*	*0.004*	*0.240*
Net Penalty	-0.005	-0.004	0.005
p-value	*0.187*	*0.274*	*0.251*
Mean Beds	0	0	0
p-value	*0*	*0*	*0*
Mean Medicaid	0	0	0
p-value	*0*	*0*	*0*
Model Fit			
# Obs	2395	2395	2395
R2	0.002	0.005	-0.002

Problem 10

Local ATE would be the average treatment effect that are only for hospitals that were penalized because of their Medicare share. This would capture the effect of penalty status on price changes only for hospitals that were penalized from Medicare patient share, making it a narrower group than all penalized hospitals. The local effect could be larger than ATE if the hospitals within the local group is more sensitive to Medicare reimbursement changes.